

Math 211 First Exam

14 October 2018

Questions 1–5: 5 points each, questions 6–10: 15 points each.

In questions 1–5, select the correct Answer. Use back of the previous page for your calculations.

1. Let $\mathbf{u}(t)$ and $\mathbf{v}(t)$ be differentiable vector-valued functions and $f(t)$ be a differentiable real-valued function.

Then $\frac{d}{dt} [f(t) [\mathbf{u}(t) \times \mathbf{v}(t)]]$ equals _____.

- (a) $f(t) [\mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)]$ (b) $f(t) [\mathbf{u}'(t) \times \mathbf{v}'(t)]$
(c) $f'(t) [\mathbf{u}'(t) \times \mathbf{v}'(t)]$ (d) $f'(t) [\mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)]$
(e) $f'(t) [\mathbf{u}(t) \times \mathbf{v}(t)] + f(t) [\mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)]$

2. The length of the curve with vector equation $\mathbf{r}(t) = \langle t^2, 2t, \ln t \rangle$, $1 \leq t \leq 3$ is _____.

- (a) $12 + \ln 3$ (b) $8 + \ln 3$ (c) $2 + \ln 3$ (d) $4 + 2 \ln 3$ (e) 2

3. Given $f(x, y) = \frac{y^2 \sin x}{x^2 + y^4}$, which of the following is false?

- (a) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the x -axis equals 0. (b) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the y -axis equals 0.
(c) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ along the line $y = x$ equals 0. (d) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.
(e) $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist.

4. Which function satisfies the equation $u_{xx} + u_{yy} = 0$?

- (a) $u(x, y) = x^2 + y^3$ (b) $u(x, y) = \sinh x \cosh y$ (c) $u(x, y) = \ln \sqrt{x^2 + y^2}$
(d) $u(x, y) = \sin x \cos y$ (e) $u(x, y) = x^3 y + 3x^2 y^2$

5. Let $F(x, y, z)$ be a differentiable function satisfying $F_x - F_y = 0$. Suppose that the equation $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y , then _____.

- (a) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$ (b) $\frac{\partial z}{\partial x} = -\frac{\partial z}{\partial y}$ (c) $\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = 1$ (d) $\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = -1$ (e) $\frac{\partial x}{\partial z} = \frac{\partial y}{\partial z}$

6. Let $f(x)$ be a function with continuous second derivative and let C be the graph $y = f(x)$.

- (a) Write a vector function that represents the curve C .
(b) Prove that the curvature at any point on C is $\kappa(x) = \frac{|y''|}{[1 + (y')^2]^{3/2}}$.

7. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2}$ exists.

8. Let $f(x, y) = x \cos y - 2^x \sin y$.

- (a) Show that f is differentiable on $\mathbb{R}^2$.
(b) Find an equation for the tangent plane to the surface $z = f(x, y)$ at the point $(1, \pi, -1)$.
(c) Use the linearization of f at $(1, \pi)$ to estimate $f(1.1, 3)$.

9. Let $f(x, y)$ be a function whose second-order partial derivatives are continuous throughout $\mathbb{R}^2$. If $z = f(x, y)$ where $x = st$, and $y = 2s + 3t$, find $\frac{\partial z}{\partial t}$ and $\frac{\partial^2 z}{\partial s \partial t}$ in terms of partial derivatives of f .

10. Find all points on the surface $x^2 - y^2 - z^2 = 4$ where the tangent plane is parallel to the plane $3x + 2y + z = 7$.

SOLUTIONS

1. The correct choice is: $f'(t)[\mathbf{u}(t) \times \mathbf{v}(t)] + f(t)[\mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)]$
2. The correct choice is: $8 + \ln 3$

$$\begin{aligned}\mathbf{r}'(t) = \langle 2t, 2, 1/t \rangle &\implies |\mathbf{r}'(t)| = \sqrt{4t^2 + 4 + 1/t^2} = (2t + 1/t) \\ &\implies L = \int_1^3 |\mathbf{r}'(t)| dt = \int_1^3 (2t + 1/t) dt = t^2 + \ln |t| \Big|_1^3 = 8 + \ln 3.\end{aligned}$$

3. The correct choice is: $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$

If $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exists and equals L , then approaching $(0, 0)$ along the x -axis we have

$$L = \lim_{x \rightarrow 0} \frac{0}{x^2} = 0,$$

while approaching $(0, 0)$ along the parabola $x = y^2$ gives

$$L = \lim_{x \rightarrow 0} \frac{x \sin x}{2x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2}.$$

This contradiction proves that the given limit does not exist.

4. The correct choice is: $u(x, y) = \ln \sqrt{x^2 + y^2}$

If $u(x, y) = \ln \sqrt{x^2 + y^2} = \frac{1}{2} \ln(x^2 + y^2)$, then $u_{xx} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$ and $u_{yy} = \frac{x^2 - y^2}{(x^2 + y^2)^2}$, hence $u_{xx} + u_{yy} = 0$.

5. The correct choice is: $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$

We have $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$ and $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$. On the other hand, it is given that $F_x = F_y$. Hence $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial y}$.

6. (a) The curve C may be parametrized as $\mathbf{r}(x) = x\mathbf{i} + f(x)\mathbf{j}$.

(b) Using the vector representation of C in (a), we have $\mathbf{r}'(x) = \mathbf{i} + f'(x)\mathbf{j}$ and $\mathbf{r}''(x) = f''(x)\mathbf{j}$. Therefore, $\mathbf{r}'(x) \times \mathbf{r}''(x) = f''(x)\mathbf{i} \times \mathbf{j} + f'(x)f''(x)\mathbf{j} \times \mathbf{j} = f''(x)\mathbf{k}$, and in turn

$$\kappa(x) = \frac{|\mathbf{r}'(x) \times \mathbf{r}''(x)|}{|\mathbf{r}'(x)|^3} = \frac{|f''(x)|}{\left(\sqrt{1 + [f'(x)]^2}\right)^3} = \frac{|y''|}{[1 + (y')^2]^{3/2}}.$$

7. Since $0 \leq y^2 \leq x^4 + y^2$, for all $(x, y) \neq (0, 0)$ we have

$$0 \leq \frac{y^2}{x^4 + y^2} \leq 1 \implies 0 \leq \frac{x^2 y^2}{x^4 + y^2} \leq x^2.$$

On the other hand, by continuity of the functions $g(x, y) = 0$ and $g(x, y) = x^2$,

$$\lim_{(x,y) \rightarrow (0,0)} 0 = \lim_{(x,y) \rightarrow (0,0)} x^2 = 0.$$

Now the Squeeze Theorem yields $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^4 + y^2} = 0$.

8. (a) The function f is differentiable everywhere in $\mathbb{R}^2$ since its partial derivatives $f_x(x, y) = \cos y - 2^x \ln 2 \sin y$ and $f_y(x, y) = -x \sin y - 2^x \cos y$ exist and are continuous throughout $\mathbb{R}^2$.
- (b) At the point $(1, \pi, -1)$, $f_x(1, \pi) = -1$ and $f_y(1, \pi) = 2$. Therefore, an equation for the desired tangent plane is

$$z + 1 = -1(x - 1) + 2(y - \pi).$$

- (c) The linearization of f at $(1, \pi)$ is $L(x, y) = -1 - 1(x - 1) + 2(y - \pi)$ using which we obtain

$$f(1.1, 3) \approx L(1.1, 3) = -1 - (0.1) + 2(3 - \pi) \approx -1 - 0.1 - 0.28 = -1.38.$$

9. Using the chain rule we obtain

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} = sf_x + 3f_y.$$

Similarly,

$$\begin{aligned} \frac{\partial^2 z}{\partial s \partial t} &= \frac{\partial}{\partial s} \left(\frac{\partial z}{\partial t} \right) = \frac{\partial}{\partial s} (sf_x + 3f_y) = f_x + s \frac{\partial}{\partial s} f_x + 3 \frac{\partial}{\partial s} f_y \\ &= f_x + s \left(\frac{\partial f_x}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f_x}{\partial y} \frac{\partial y}{\partial s} \right) + 3 \left(\frac{\partial f_y}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial f_y}{\partial y} \frac{\partial y}{\partial s} \right) \\ &= f_x + s(tf_{xx} + 2f_{xy}) + 3(tf_{yx} + 2f_{yy}). \end{aligned}$$

Since the second-order partial derivatives of f are continuous throughout $\mathbb{R}^2$, we have $f_{xy} = f_{yx}$ by Clairaut's Theorem. Therefore,

$$\frac{\partial^2 z}{\partial s \partial t} = f_x + stf_{xx} + (2s + 3t)f_{xy} + 6f_{yy} \quad (= f_x + xf_{xx} + yf_{xy} + 6f_{yy}).$$

10. Let $F(x, y, z) = x^2 - y^2 - z^2$. Since the tangent plane at any point (a, b, c) on the surface $F(x, y, z) = 4$, is normal to the gradient vector $\nabla F(a, b, c) = \langle 2a, -2b, -2c \rangle$, the tangent is parallel to the plane $3x + 2y + z = 7$, if and only if $\langle 2a, -2b, -2c \rangle = k\langle 3, 2, 1 \rangle$ for some $k \in \mathbb{R}$. That is, if $a = 3k/2$, $b = -k$, and $c = -k/2$. On the other hand, since (a, b, c) belongs to the surface $F(x, y, z) = 4$,

$$a^2 - b^2 - c^2 = 4 \iff \frac{9k^2}{4} - k^2 - \frac{k^2}{4} = 4 \iff k^2 = 4 \iff k = \pm 2.$$

In conclusion, the desired points are $(3, -2, -1)$ and $(-3, 2, 1)$.